

Введение в БСППР

1 СППВР

Clinical Decision Support System (CDSS)

Биомедицинская система поддержки принятия решений (БСППР)

Система поддержки принятия врачебных решений (СППВР)

Clinical decision support systems (CDSS) are computer systems designed to impact clinician decision making about individual patients at the point in time that these decisions are made.

Система поддержки принятия врачебных решений (СППВР) — это программное обеспечение, позволяющее путем сбора и анализа информации влиять на принятие врачом решения при обследовании пациента, диагностике, назначении лечения с целью снижения ошибок и повышения качества оказываемой медицинской помощи.

1 СППВР

1.1 Типы СППВР

1.1.1 Knowledge-Based Clinical Decision Support Systems (СППВР, основанная на знаниях; «знаниевые» СППВР)

Правила, экспертные знания

1.1.2 Nonknowledge-Based Clinical Decision Support Systems («незнаниевые» СППВР)

Машинное обучение, большие данные

2 Математические основы систем поддержки принятия решений

2.1 Обзор подходов

2.1.1 Теория множеств

$$\text{JONES} - \text{CRITERIA} - \text{MAJOR} = \left\{ \begin{array}{l} \text{carditis, migratory polyarthritis,} \\ \text{erythema marginatum, chorea,} \\ \text{subcutaneous nodules} \end{array} \right\}$$

$$\text{JONES} - \text{CRITERIA} - \text{MINOR} = \left\{ \begin{array}{l} \text{fever, arthralgia, elevated acute phase reactants,} \\ \text{prolonged P - R interval on electrocardiogram} \end{array} \right\}$$

$$\text{GROUP} - \text{A} - \text{STREP} - \text{EVIDENCE} = \left\{ \begin{array}{l} \text{positive culture, positive rapid antigen,} \\ \text{antibody rise or elevation} \end{array} \right\}$$

$$\text{FINDINGS} = \{ \text{migratory polyarthritis, chorea, subcutaneous nodules} \}$$

$$\text{FINDINGS} \subseteq \text{JONES} - \text{CRITERIA} - \text{MAJOR}$$

$$|\text{FINDINGS}| = 3$$

$$|\text{JONES} - \text{CRITERIA} - \text{MAJOR}| = 5$$

$$\text{CLINICAL} - \text{FINDINGS} = \left\{ \begin{array}{l} \text{heart murmur,} \\ \text{migratory polyarthritis,} \\ \text{chorea, subcutaneous nodules,} \\ \text{cough} \end{array} \right\}$$

2 Математические основы систем поддержки принятия решений

2.1 Обзор подходов

2.1.1 Теория множеств

CLINICAL – FINDINGS $\cap$ JONES – CRITERIA – MAJOR

CLINICAL – FINDINGS $\cup$ JONES – CRITERIA – MAJOR

$= \left\{ \begin{array}{l} \text{heart murmur, migratory polyarthritis, chorea, subcutaneous} \\ \text{nodules, cough, carditis, erythema marginatum, chorea} \end{array} \right\}$

2 Математические основы систем поддержки принятия решений

2.1 Обзор подходов

2.1.1 Теория множеств

$$|\text{CLINICAL} - \text{FINDINGS} \cap \text{JONES} - \text{CRITERIA} - \text{MAJOR}| \geq 2 \quad (2.1)$$

and

$$|\text{CLINICAL} - \text{FINDINGS} \cap \text{GROUP} - \text{A} - \text{STREP} - \text{EVIDENCE}| \geq 1$$

$$|\text{CLINICAL} - \text{FINDINGS} \cap \text{JONES} - \text{CRITERIA} - \text{MAJOR}| = 1 \quad (2.2)$$

and

$$|\text{CLINICAL} - \text{FINDINGS} \cap \text{JONES} - \text{CRITERIA} - \text{MINOR}| \geq 2$$

and

$$|\text{CLINICAL} - \text{FINDINGS} \cap \text{GROUP} - \text{A} - \text{STREP} - \text{EVIDENCE}| \geq 1$$

2 Математические основы систем поддержки принятия решений

2.1 Обзор подходов

2.1.1 Теория множеств

$\text{PATHOGENS} = \{A, B, C, D, E, F\}$

$\text{ANTIBIOTICS} = \{A - \text{Cillin}, B - \text{Cillin}, C - \text{Cillin}, D - \text{Cillin}, E - \text{Cillin}, F - \text{Cillin}\}$

$A - \text{Cillin} = \{A, C\}$

$B - \text{Cillin} = \{A, B, E\}$

$C - \text{Cillin} = \{C, D, E\}$

$D - \text{Cillin} = \{F\}$

$E - \text{Cillin} = \{B, D, F\}$

$F - \text{Cillin} = \{E\}$

Set Covering

2 Математические основы систем поддержки принятия решений

2.1 Обзор подходов

2.1.2 Булева логика

Major Premise : Low – CO₂ $\Rightarrow$ OVERVENTILATED

Minor Premise : Low – CO₂

Conclusion : OVERVENTILATED

Major Premise : Low – CO₂ $\Rightarrow$ OVERVENTILATED

Minor Premise : OVERVENTILATED $\Rightarrow$ HIGH – RATE – ALARM

Conclusion : Low – CO₂ $\Rightarrow$ HIGH – RATE – ALARM

2 Математические основы систем поддержки принятия решений

2.1 Обзор подходов

2.1.2 Булева логика

Forward-chaining deduction system

Backward-chaining deduction system

The question a clinician might ask is not “Does this patient have asthma? If so, I should give a flu shot,” but more likely the question would be simply “Does this patient need a flu shot?” We start with the “then” portion of this set of flu shot rules. A backward-chaining deduction system does this — it starts with the “then” end of a set of rules and works backwards to answer questions based on its rule set. In the flu shot example, a backward-chaining system would start with the “Does this patient need a flu shot” question and immediately learn that the diagnosis of asthma would cause this rule to be satisfied.

2 Математические основы систем поддержки принятия решений

2.1 Обзор подходов

2.1.3 Вероятность

$$\Pr(\text{diabetes}) = 0.6$$

$$\Pr(\text{hypertension}) = 0.3$$

$$\Pr(\text{diabetes AND hypertension}) = 0.6 \times 0.3 = 0.18$$

$$\begin{aligned} \Pr(\text{diabetes OR hypertension}) &= \Pr(\text{diabetes}) + \Pr(\text{hypertension}) \\ &\quad - \Pr(\text{diabetes AND hypertension}) \end{aligned}$$

$$\Pr(\text{diabetes OR hypertension}) = 0.6 + 0.3 - 0.18 = 0.72.$$

2 Математические основы систем поддержки принятия решений

2.1 Обзор подходов

2.1.4 Теорема Байеса

$$\frac{\Pr(\text{hepatitis} \mid \text{jaundice})}{\Pr(\text{jaundice} \mid \text{hepatitis})}$$

$$\Pr(\text{hepatitis} \mid \text{jaundice}) = \frac{\Pr(\text{hepatitis}) \times \Pr(\text{jaundice} \mid \text{hepatitis})}{\Pr(\text{jaundice})}$$

2 Математические основы систем поддержки принятия решений

2.1 Обзор подходов

2.1.4 Теорема Байеса

$$\begin{aligned}\Pr(\text{diabetes AND hypertension}) &= \Pr(\text{diabetes}) \times \Pr(\text{hypertension}) \\ &= 0.6 \times 0.3 = 0.18.\end{aligned}$$

$$\begin{aligned}\Pr(\text{hypertension}) &= \Pr(\text{hypertension} \mid \text{diabetes}) \text{ and} \\ \Pr(\text{diabetes}) &= \Pr(\text{diabetes} \mid \text{hypertension}).\end{aligned}$$

2 Математические основы систем поддержки принятия решений

2.1 Обзор подходов

2.1.4 Теорема Байеса

$\Pr(\text{pneumonia AND abnormal CXR})$

WRONG: $\Pr(\text{pneumonia AND abnormal CXR}) = 1 / 1000 \times 1 / 100 =$
 $0.00001 = 0.001\%$

$\Pr(\text{pneumonia AND abnormal CXR}) = \Pr(\text{pneumonia}) \times \Pr\left(\begin{matrix} \text{abnormal CXR} \\ \text{pneumonia} \end{matrix}\right)$

2 Математические основы систем поддержки принятия решений

2.1 Обзор подходов

2.1.4 Теорема Байеса

$$\Pr(A \text{ AND } B) = \Pr(A) \times \Pr(B \mid A) \text{ and}$$

$$\Pr(B \text{ AND } A) = \Pr(B) \times \Pr(A \mid B).$$

$$\Pr(A) \times \Pr(B \mid A) = \Pr(B) \times \Pr(A \mid B)$$

$$\Pr(A \mid B) = \frac{\Pr(A) \times \Pr(B \mid A)}{\Pr(B)}$$

2 Математические основы систем поддержки принятия решений

2.1 Обзор подходов

2.1.5 Неформальная логика

MYCIN (1970s at Stanford University)

early backward chaining expert system that used artificial intelligence to identify bacteria causing severe infections, such as bacteremia and meningitis, and to recommend antibiotics

own logic system (heuristic) that made intuitive sense

“certainty factors” which ranged from -1 (false) to $+1$ (true)

3 Data Mining и СППВР

Data mining == интеллектуальный анализ данных

“Data mining is a process of pattern and relationship discovery within large sets of data.”

“Data mining is the process of extracting and finding patterns in massive data sets involving methods at the intersection of machine learning, statistics, and database systems.”

3 Data Mining и СППВР

Обучение с учителем (Supervised Learning)

Обучение без учителя (Unsupervised Learning)

Обучение с частичным привлечением учителя (Semi-supervised Learning)

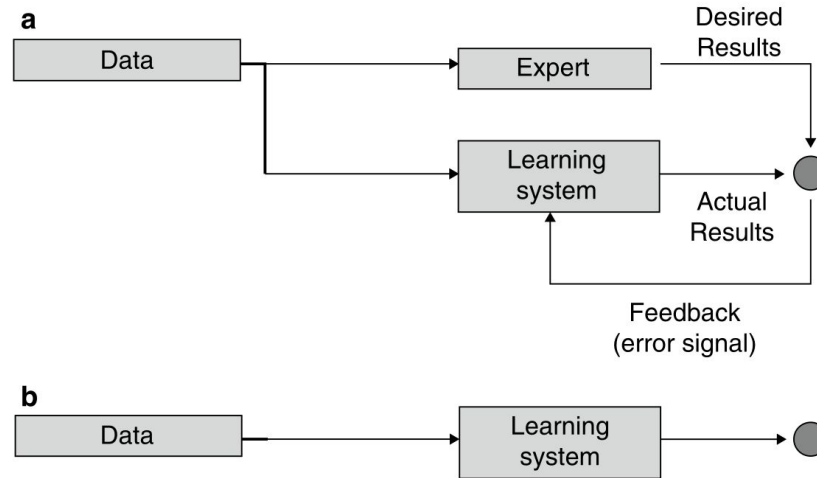
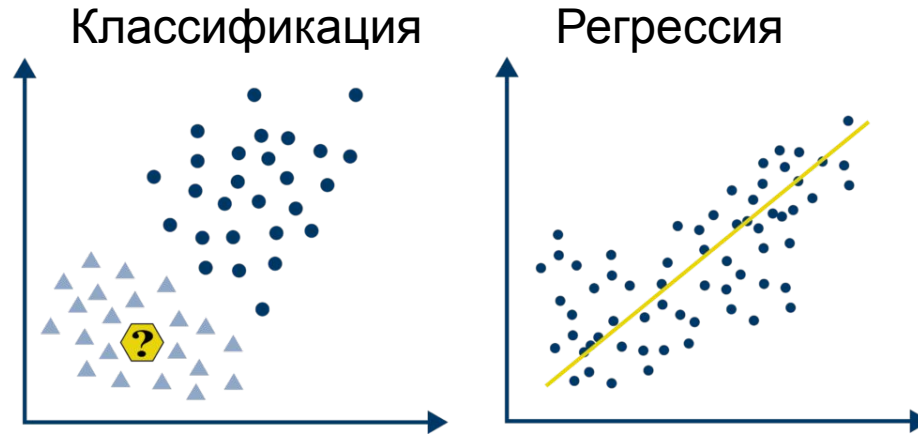


Fig. 3.1 (a) Supervised learning; (b) unsupervised learning

3 Data Mining и СППВР

3.1 Обучение с учителем (Supervised Learning)



3 Data Mining и СППВР

3.1.1 Классификаторы для обучения с учителем

Дерево решений (Decision Trees)

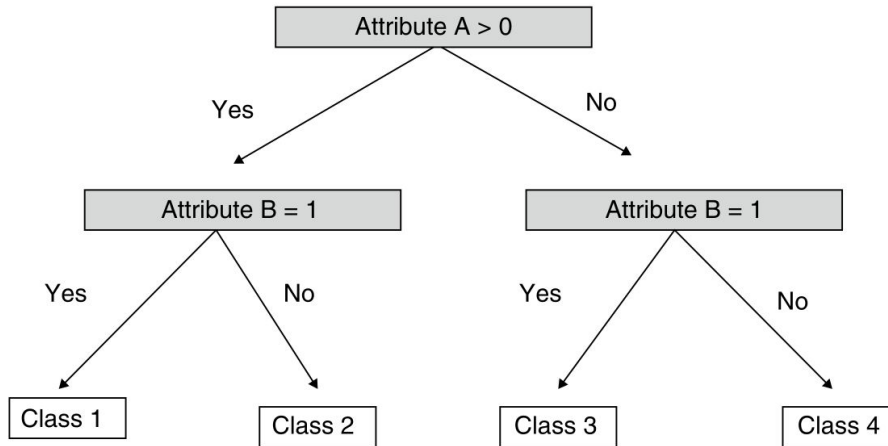


Fig. 3.2 A simple decision tree with the tests on attributes A and B

Метод случайного леса (Random Forests, RF)

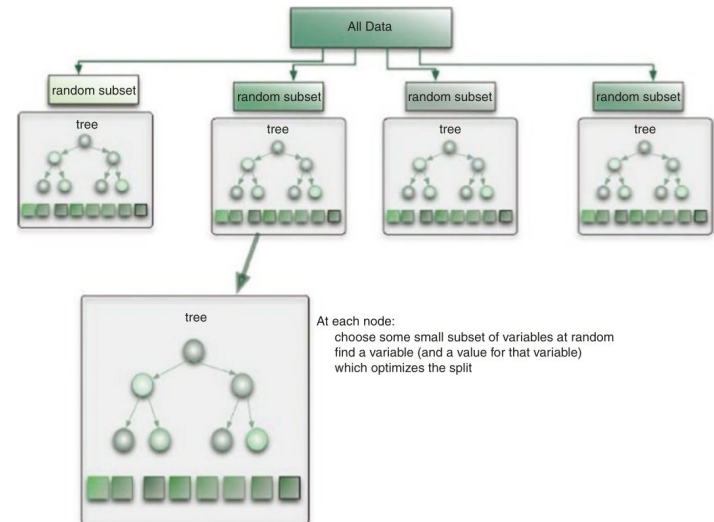


Fig. 9.2 Sketch representation of a RF workflow. RFs trained different algorithms by looking at random subsets of the data. The randomness generates models that are not correlated to each other

3 Data Mining и СППВР

3.1.1 Классификаторы для обучения с учителем

Метод k-ближайших соседей (k-Nearest Neighbor Classifier)

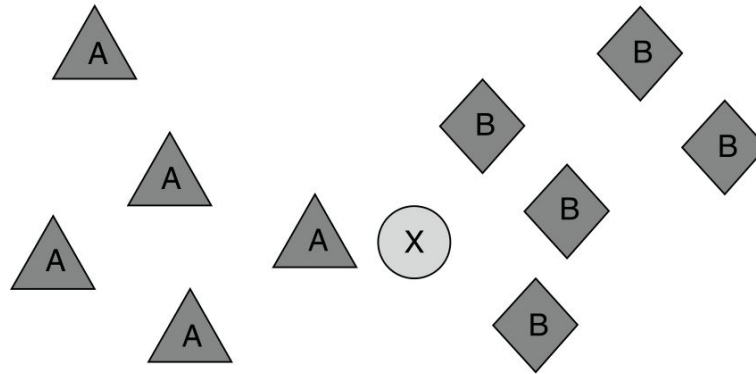


Fig. 3.5 Nearest neighbor (NN) classifier. There are two classes, A (*triangles*) and B (*diamonds*). The *circle* represents the unknown sample, X. For the NN rule the nearest neighbor of X comes from class A, so it would be labeled class A. Using the k-NN rule with $k=4$, three of the nearest neighbors of sample X come from class B, so it would be labeled as B

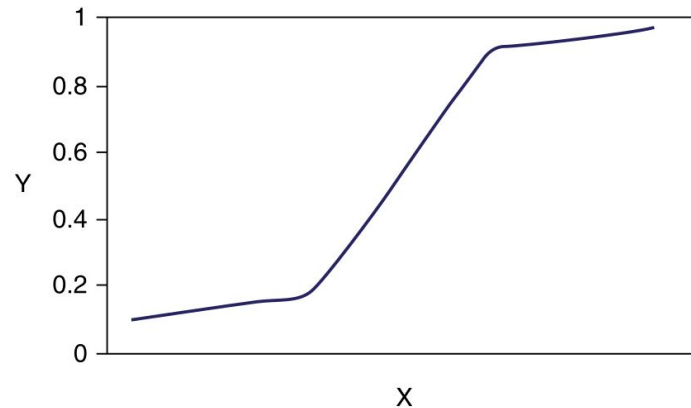
3 Data Mining и СППВР

3.1.1 Классификаторы для обучения с учителем

Логистическая регрессия (Logistic Regression)

$$p(y) = 1 / \{1 + e^{(-a - bx)}\}$$

Fig. 3.3 Logistic regression model



3 Data Mining и СППВР

3.1.1 Классификаторы для обучения с учителем

Метод опорных векторов (Support Vector Machines, SVMs)

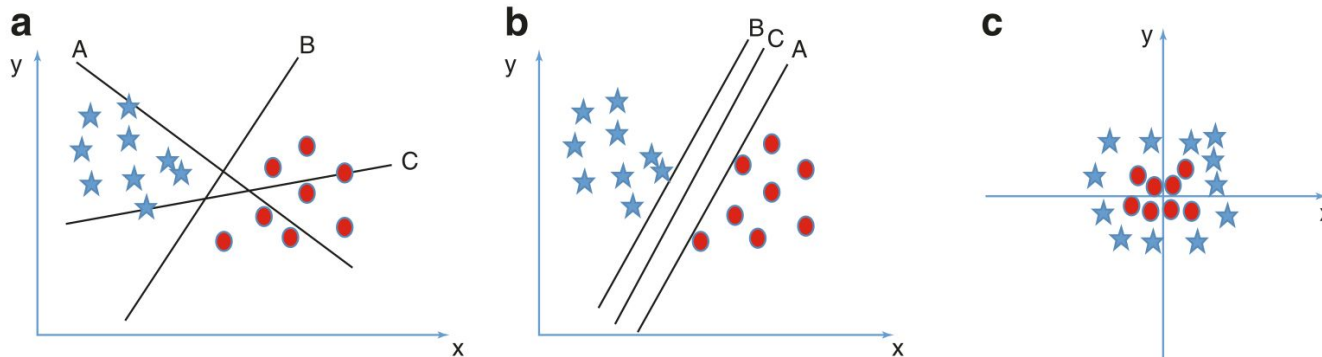


Fig. 9.1 SVMs examples. **(a)** Three different solutions of the problems are drawn. The solution that optimizes the separation between the two clusters of data (stars vs circles) is line B. **(b)** the optimal solution is line C, by keeping into consideration the concept of margin. However, non linear solutions might be needed as shown in **(c)**

3 Data Mining и СППВР

3.1.1 Классификаторы для обучения с учителем

Нейронные сети (Neural Networks)

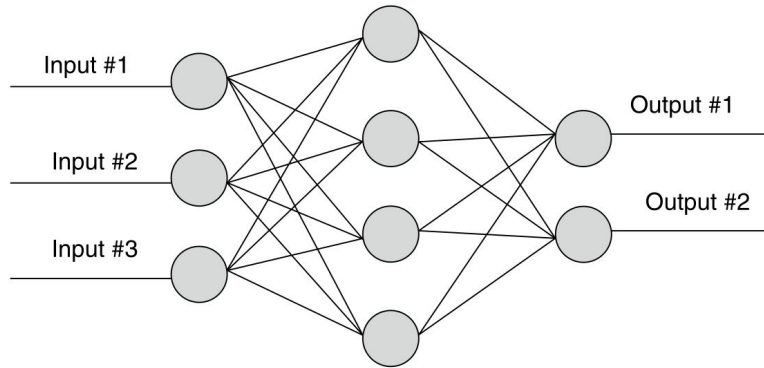
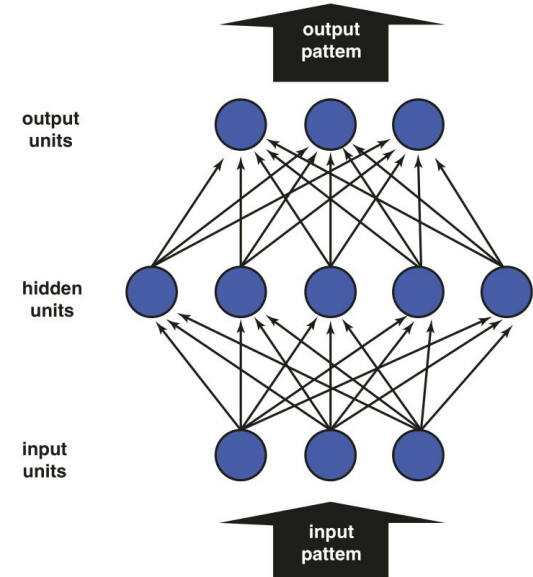


Fig. 3.4 Neural network

Fig. 9.3 Example architecture for an ANN. The standard architecture has one input layer, output units, and an intermediate layer called 'hidden units'



3 Data Mining и СППВР

3.1.2 Оценка качества работы модели

Оценка качества регрессии

$MSE = \frac{1}{n} \sum_{i=1}^n (a(x_i) - y_i)^2$	Средняя квадратичная ошибка (англ. Mean Squared Error, MSE)
$MAE = \frac{1}{n} \sum_{i=1}^n a(x_i) - y_i $	Средняя абсолютная ошибка (англ. Mean Absolute Error, MAE)
$R^2 = 1 - \frac{\sum_{i=1}^n (a(x_i) - y_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$	Коэффициент детерминации
$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (a(x_i) - y_i)^2}$	Корень из средней квадратичной ошибки (англ. Root Mean Squared Error, RMSE)
$MAPE = 100\% \times \frac{1}{n} \sum_{i=1}^n \frac{ y_i - a(x_i) }{ y_i }$	Средняя абсолютная процентная ошибка (англ. Mean Absolute Percentage Error, MAPE)

3 Data Mining и СППВР

3.1.2 Оценка качества работы модели Матрица ошибок (Confusion Matrix)

Table 8.4 Confusion matrix showing predictions and observations. Many useful performance metrics are derived from the values in the confusion matrix

		Observation		
		True	False	
Prediction	True	True positive (TP)	False positive (FP)	→ Positive predictive value (PPV)
	False	False negative (FN)	True negative (TN)	→ Negative predictive value (NPV)
		↓	↓	
		Sensitivity (TPR)	Specificity (TNR)	

https://en.wikipedia.org/wiki/Precision_and_recall

3 Data Mining и СППВР

3.1.2 Оценка качества работы модели

$$\text{Accuracy} = (TN + TP) / (TN + TP + FP + FN)$$

$$\text{PPV (Positive Predictive Value, precision)} = TP / (TP + FP)$$

$$\text{NPV (Negative Predictive Value)} = TN / (TN + FN)$$

$$\text{TPR (True Positive Rate, recall, sensitivity)} = TP / (TP + FN)$$

$$\text{TNR (True Negative Rate, specificity)} = TN / (TN + FP)$$

$$\text{FPR (False Positive Rate)} = 1 - \text{TPR} = 1 - \text{sensitivity}$$

$$\text{FNR (False Negative Rate)} = 1 - \text{TNR} = 1 - \text{specificity}$$

$$\text{F1-score} = 2 \cdot (\text{PPV} \cdot \text{TPR}) / (\text{PPV} + \text{TPR})$$

3 Data Mining и СППВР

3.1.2 Оценка качества работы модели

ROC (Receiver Operating Characteristic) кривая

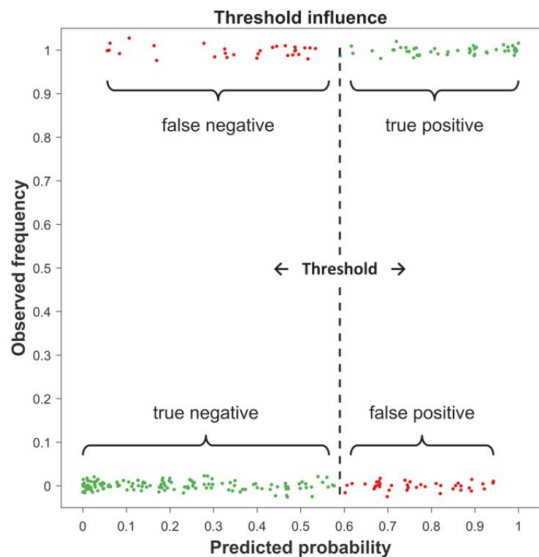


Fig. 8.4 Influence of the threshold that is used to round model prediction probabilities to 0 or 1. By using a low threshold the model will detect most of the patients with the outcome (high sensitivity), but many patients without the outcome will also be included (low specificity). For each value of the threshold sensitivity and specificity values can be calculated

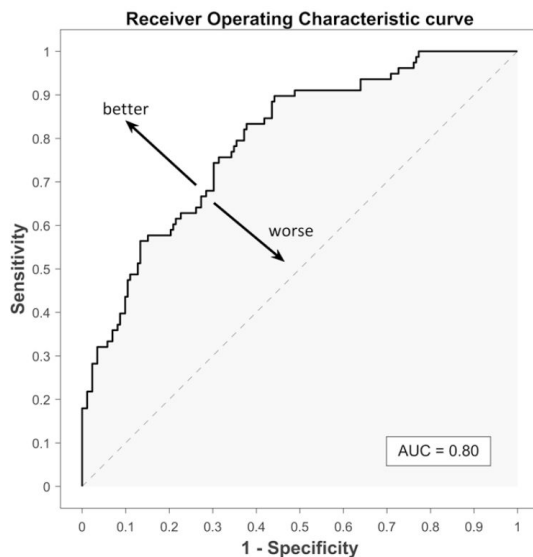
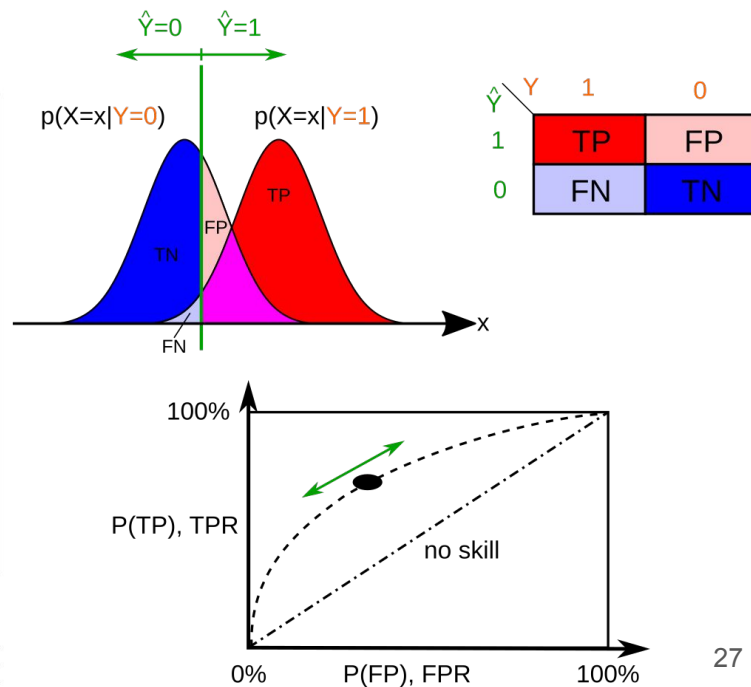


Fig. 8.5 ROC curve indicating discriminating performance of the model. Model predictions are rounded to 0 or 1 using many different thresholds resulting in the sensitivity and specificity pairs that form the curve. AUC is indicated by the gray area under the curve. Higher values correspond to better model discrimination performance

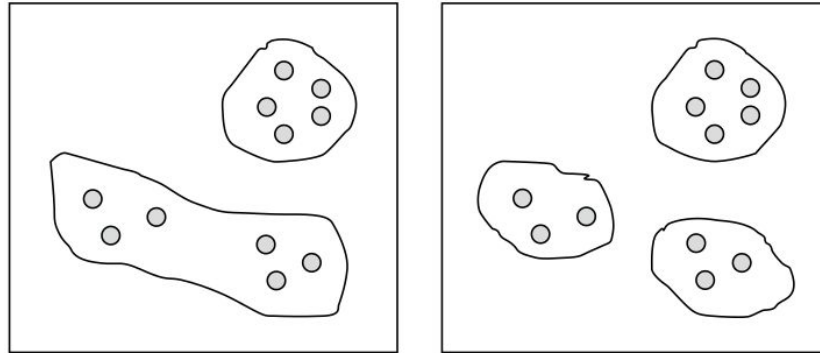


3 Data Mining и СППВР

3.2 Обучение без учителя (Unsupervised Learning)

3.2.1 Кластерный анализ (Cluster Analysis)

Fig. 3.7 Cluster analysis.
Two clusters of data (*left*);
three clusters (*right*) using
the same set of data



3 Data Mining и СППВР

3.2 Обучение без учителя (Unsupervised Learning)

3.2.1 Кластерный анализ (Cluster Analysis)

K-means

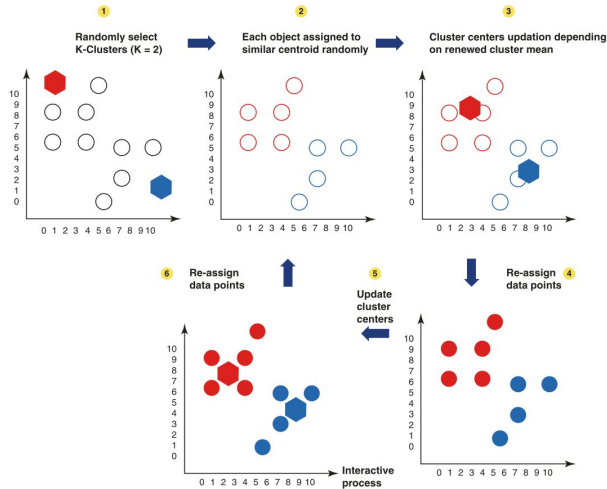
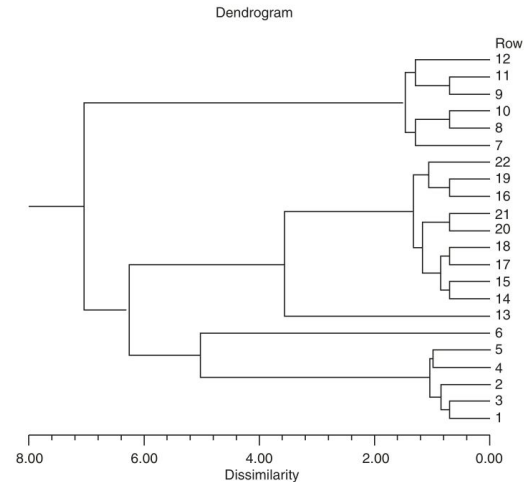


Fig. 9.4 Sketch of the k-means clustering algorithm

Иерархическая кластеризация (Hierarchical Clustering)

Fig. 9.5 Example of a dendrogram



3 Data Mining и СППВР

3.3 Другие методы

3.3.1 Генетические алгоритмы (Genetic Algorithms)

3.3.2 Биологические вычисления (Biological Computing)

3.3.3 Квантовые вычисления (Quantum Computing)

3.3.4 Внедрение нечеткой логики и других гибридных методов (Incorporating Fuzzy Logic and Other Hybrid Methods)

3.3.5 Аналитика больших данных (Big Data Analytics)

4 Нечеткая логика (Fuzzy Logic)

<https://www.mathworks.com/help/fuzzy/foundations-of-fuzzy-logic.html>

<https://www.mathworks.com/help/fuzzy/fuzzy-inference-process.html>

<https://www.mathworks.com/help/fuzzy/types-of-fuzzy-inference-systems.html>

4 Резюме

1 Введение в СППВР

2 Математические основы систем поддержки принятия решений,
вычислительная диагностика

3 Data Mining и СППВР (введение в методы машинного обучения)

4 Введение в нечеткую логику

Материалы

Спиридонов И.Н., А. В. Самородов А.В. Методы и алгоритмы вычислительной диагностики

<https://press.bmstu.ru/catalog/item/2627/>